

Micro-Transformer: A From-Scratch Autograd and Attention Engine for Character-Level Language Modeling

Team Members: Manikanta Harsha Nadimpalli, Luis Blanco Hoyos

Emails (SUNet): nmharsha@stanford.edu, blancol@stanford.edu

1 Motivation

We are building a reverse-mode autograd engine and a small transformer [1] on top of it from scratch in NumPy, with one specific scientific question in mind: *how do LayerNorm [4] placement (pre-norm, post-norm, none) and residual connections shape the gradient signal that reaches the early layers of a decoder-only transformer during training on character-level language modeling?* The from-scratch build is the instrument; the gradient-flow ablation is the contribution. Existing frameworks expose hooks and functional Jacobians, but constructing the engine ourselves gives us conceptual clarity over every chain-rule edge and exact, low-overhead access to per-layer activation and gradient statistics, with no framework internals to reverse-engineer.

The milestone deliverables described below are the foundation of that instrument: a data pipeline, three classical baselines, a scalar reverse-mode engine, a tensor reverse-mode engine with the full op surface needed for an MLP, and a first non-trivial language model trained end-to-end through our own engine. The ablation grid itself is post-milestone work and tracked under *Next Steps*.

2 Methods

2.1 Data and tokenization

Tiny Shakespeare [3] (1,115,394 characters, 65 unique symbols), char-level tokenization. Train/val split is 90/10 *by position*, not random; random char-level shuffling would leak local context across the boundary and inflate val performance. Sampling uses caller-supplied `np.random.Generator` so every run is reproducible at the seed.

2.2 Baselines (closed-form and SGD)

Three reference points anchor every later result:

- **Uniform predictor.** Cross-entropy floor of $\log V$ nats.
- **Closed-form bigram** with Laplace smoothing ($\alpha=1$). Direct count $\rightarrow$ normalize on train. Smoothing is required because val contains bigrams unseen in train.
- **Softmax regression by SGD.** The same model class as the bigram (logits = $W[i] + b$) trained by mini-batch SGD with hand-derived gradients in plain NumPy. Convergence behavior under SGD is itself a sanity check: the SGD trajectory should approach the closed-form CE.

2.3 Scalar reverse-mode autograd

A Karpathy-style [2] `Value` class wrapping a Python float. Eight elementary ops (`+`, `*`, `**`, `relu`, `tanh`, `exp`, `log`, `neg/sub/div` as compositions). Each forward op installs a closure that captures (`self`, `other`, `out`) and pushes the upstream gradient through its local Jacobian; `backward()` topologically sorts from the output and walks in reverse. Verified against (i) central differences and (ii) `torch.autograd` to bit-identical agreement on a randomized two-layer neuron.

2.4 Tensor reverse-mode autograd

A NumPy-backed `Tensor` class lifting the same closure pattern to batched arrays. The two genuine complications are broadcasting in backward (collapsed by an `unbroadcast` helper that sums along broadcast leading axes and size-1 dims) and lazy grad allocation. The

op surface is sufficient for the MLP and forward-compatible with attention: elementwise binary/unary with broadcasting, reductions `sum/mean` (axis + keepdims) with grad re-broadcast, batched matmul (`@`) with leading-batch fan-in, integer-array gather (the embedding lookup, with `np.add.at` scatter-add backward), reshape, `relu`, `tanh`, numerically stable `log_softmax`, and a fused `cross_entropy(logits, targets)` whose backward is the closed-form $(\text{softmax} - \text{onehot})/B$. Every op has a numerical-gradient-check test (the non-negotiable Week 2 deliverable from our project plan).

2.5 First model: char-MLP via our engine

A Bengio-style [5] character-MLP: previous $T=3$ characters $\rightarrow$ embedding $\mathbb{R}^{V \times d} \rightarrow$ flatten $\rightarrow$ `tanh`-hidden $\rightarrow$ linear $\rightarrow$ cross-entropy on the next character. 8,401 trainable parameters; trained end-to-end with our autograd, no torch in the loop.

2.6 Closed-form attention backward (parallel work, Week-4 spec)

A complete derivation of the scaled-dot-product attention backward — softmax-row Jacobian collapse, batched-matmul fan-in, three projection paths summing into X , causal-mask handling — together with a PyTorch reference implementation and four parity tests at `float64`, $1\text{e-}9$ tolerance against `torch.autograd.grad`. This is the spec we will port into the Tensor engine in the next phase.

3 Preliminary Experiments and Results

3.1 Headline

Table 1 reports validation cross-entropy on Tiny Shakespeare across all four models. The MLP, trained end-to-end through our autograd engine, beats the closed-form bigram floor by 0.20 nats (0.29 bpc, an 18% reduction in perplexity).

Model	Val CE (nats)	Val bpc	Val perplexity
Uniform	4.1744	6.0224	65.00
Softmax regression (SGD, 3k steps)	2.6733	3.8568	14.49
Bigram (closed form, $\alpha = 1$)	2.4819	3.5806	11.96
Char-MLP via our autograd	2.2795	3.2886	9.77

Table 1: Validation cross-entropy on Tiny Shakespeare (111,539 chars). Lower is better; bpc = nats / $\ln 2$. The MLP uses 3-character context and 8,401 parameters; trained for 4,000 SGD steps in 14 seconds (numpy-only, single CPU).

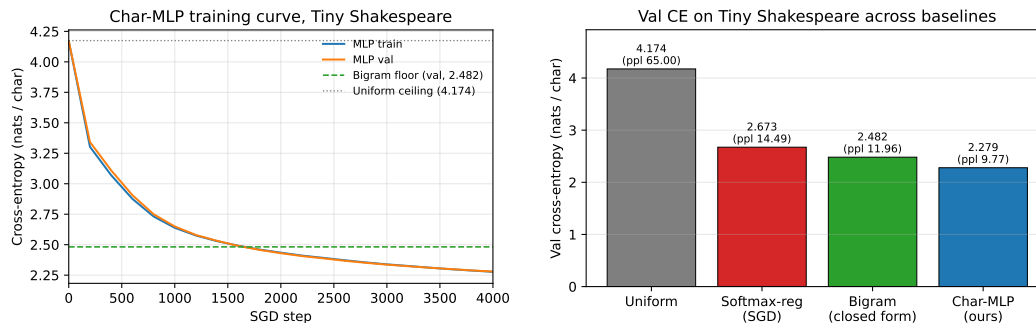
3.2 Why the MLP wins, and by how much we expected

The bigram is the optimal 2-gram model for this data; SGD on the same parameterization (table softmax-regression) cannot beat it asymptotically. The MLP sits in a strictly larger hypothesis class — it conditions on 3 characters of context and learns a non-linear feature combination via the `tanh` hidden layer. A 0.20-nat val improvement is in line with what 3-gram-style models achieve on this corpus and confirms the engine’s gradients are propagating useful learning signal, not just numerically correct noise.

3.3 Engine validation

Correctness, not raw performance, is the immediate concern: an undetected gradient bug invalidates every later experiment. We layered three checks:

- **Numerical gradient check** on every op individually (*add*, *mul*, *div*, *pow*, broadcast cases, reductions on every keepdims/axis combo, matmul on three batching patterns, gather with repeated indices, reshape, `tanh`, `relu`, `log_softmax`, `cross_entropy`). Tolerance `atol=1e-7`, `rtol=1e-5`; all 59 cases pass.



(a) MLP train+val CE vs. SGD step. Val crosses the bigram floor (dashed green) around step 1700. (b) Val CE across the four reference points. Lower is better.

Figure 1: Headline results from the engine and baselines.

- **End-to-end gradcheck on the MLP forward pass:** gradient of the CE loss against all five parameter tensors (E, W_1, b_1, W_2, b_2) on randomized inputs.
- **PyTorch parity:** same MLP forward and backward computed in our engine and in `torch.autograd`; loss bit-identical, all five gradients within 5×10^{-17} — below float64 machine epsilon ($\sim 2.2 \times 10^{-16}$).

The total project test suite has 103 passing cases.

3.4 What the eval pipeline gives us for free

Because `full_corpus_ce` runs over every consecutive bigram pair (rather than a sampled subset), val CE is directly comparable across model classes — bigram counts, SGD-trained logits, and MLP forward all consume the same sequence and report the same mean over the same 111,539 positions. The -0.20 -nat gap in Table 1 reflects model class, not evaluation noise.

4 Next Steps

1. **Port Luis's closed-form attention backward into the Tensor engine.** The derivation, PyTorch reference, and $1e-9$ -tolerance parity tests already exist (committed under `tests/reference_attention.py`); the engine has `matmul`, `log-softmax`, and a gather-shaped scatter, which is everything needed. Estimated 2–3 days.
2. **LayerNorm forward + backward**, validated by gradcheck and torch parity.
3. **Decoder block** with toggleable LayerNorm placement (pre-norm / post-norm / none) and residual on/off — exactly the 6 configurations the ablation grid will sweep.
4. **Ablation grid.** Train all 6 configs on Tiny Shakespeare, single seed, identical batches (the data layer's reproducible RNG was built for this). Per-step log: layer-wise $\|g\|_2$, activation norms, and the singular-value spectrum of layer Jacobians.
5. **Comparison curves:** gradient-norm trajectories per config, training curves per config, val perplexity at convergence.

The MLP result here is a sanity floor; the transformer ablation is the contribution.

5 Team Contributions

- **Manikanta Harsha Nadimpalli:** Tiny Shakespeare data pipeline (loader, tokenizer, batched sampler with reproducibility); three classical baselines (uniform, closed-form Laplace bigram, softmax-regression by SGD with hand-derived gradients); scalar reverse-mode autograd engine with numerical gradient check + PyTorch

parity; tensor reverse-mode autograd engine (broadcasting, reductions, matmul, gather, reshape, activations, log-softmax, cross-entropy) with per-op numerical gradient checks and end-to-end PyTorch parity; char-MLP forward and training loop; experiment scripts, results JSONs, figures; this writeup.

- **Luis Blanco Hoyos:** Closed-form derivation of the scaled-dot-product attention backward (softmax-row Jacobian collapse, batched-matmul fan-in, three-path X contribution, causal-mask handling) in `wroteups/cs229/attention_backward.md`; PyTorch reference forward and closed-form backward in `tests/reference_attention.py`; four `float64` parity tests at `1e-9` tolerance against `torch.autograd`, with and without causal mask. This is the exact specification the Week-4 attention port will be built and tested against.

References

- [1] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I. (2017). Attention is all you need. *NeurIPS* 30.
- [2] Karpathy, A. (2020). micrograd: A tiny scalar-valued autograd engine. <https://github.com/karpathy/micrograd>.
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- [4] Ba, J.L., Kiros, J.R., Hinton, G.E. (2016). Layer Normalization. *arXiv:1607.06450*.
- [5] Bengio, Y., Ducharme, R., Vincent, P., Jauvin, C. (2003). A Neural Probabilistic Language Model. *JMLR* 3, 1137–1155.